Kirti Pratihar

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Summary

Aspiring AI/ML Engineer | B.Tech Computer Science (AI & ML) Student Disciplined problem solver with a deep foundation in Data Structures & Algorithms (500+ LeetCode problems, 500-day streak). Hands-on experience architecting scalable Generative AI systems (RAG pipelines), optimizing deep learning models, and contributing to high-impact open-source infrastructure. Eager to apply advanced machine learning frameworks and algorithmic optimization in an enterprise environment.

Technical Skills

- **Programming Languages**: Python, C++, Java, SQL
- **AI Machine Learning**: LLMs, RAG Architectures, Vector Databases (Pinecone), Neural Networks, CNNs, Transformers, TensorFlow, PyTorch, Scikit-learn, LangChain, NLP
- **Cloud Tools**: AWS, Microsoft Azure (DP-900), Git, GitHub, Docker, Postman

Work Experience

Technical Lead, Matrix Club, *VIT Bhopal University*
Jan 2025 - Present

- Led technical operations for the 'Escape the Matrix' event, managing the digital registration infrastructure for 50+ participating teams.
- Served as the primary technical point of contact, ensuring seamless digital communication and support for club members.

Technical Training, *India Space Lab Winter Internship*
Dec 2024 - Jan 2025

- Executed advanced drone flight operations and analyzed space technologies including CubeSat integration and CanSat deployment.

Projects

Generative AI RAG Chat Assistant (YAP ENGINE) GitHub Link

- Architected a modular Retrieval-Augmented Generation (RAG) system processing complex document structures using LangChain and Python.
- Optimized semantic indexing by deploying a high-density vector space pipeline in Pinecone, reducing retrieval latency for context-aware queries.
- Integrated the Groq API for low-latency LLM inference and Engineered the response pipeline to reduce AI hallucinations by anchoring LLM outputs in retrieved vector data.

Credit Card Approval Prediction System GitHub Link

- Developed an end-to-end predictive machine learning pipeline utilizing a Random Forest Classifier to achieve 90% accuracy.
- Addressed severe class imbalance by implementing SMOTE (Synthetic Minority Over-sampling Technique), boosting minority class recall and model fairness.
- Engineered automated feature extraction layers to capture non-linear relationships in financial demographic data.

Taxi Trip Duration Prediction Analysis GitHub Link

- Optimized an XGBoost model to predict taxi trip durations, achieving a Root Mean Squared Log Error (RMSLE) of **0.3151**.
- Conducted spatial-temporal feature engineering by applying K-Means clustering to geographic coordinate data, uncovering dense traffic patterns.

Education

VIT Bhopal University, *B.Tech in Computer Science (AI & Machine Learning) | CGPA: 8.88*
Bhopal, India — *Expected 2027*
Kendriya Vidyalaya Bamangachi, *XII (CBSE-2023) - 84.4%*
Kendriya Vidyalaya Bamangachi, *X (CBSE-2021) – 95.2%*

Achievements & Certifications

- Awarded 1st Prize at InnovMinds Expo Hackathon (Feb 2025) for a dementia support app utilizing AI classification.
- **LeetCode Profile**: Solved 500+ Data Structures & Algorithms problems; maintained 500-day consistency streak.
- Ranked 79 in India Space Lab Internship Exam (Merit-based national competition).
- **Microsoft Certified**: Azure Data Fundamentals (DP-900).
- Advanced Learning Algorithms - Stanford University (May 2025).
- Introduction to TensorFlow for AI, Machine Learning, and Deep Learning - DeepLearning.AI (March 2025).